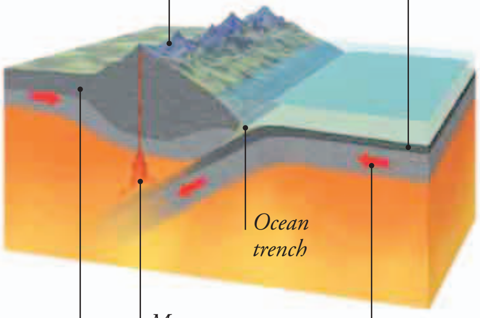
The deepest depths

In some parts of the ocean floor, titanic forces within the planet are pulling the plates of Earth's crust apart. In other places, the same forces are pushing plates together. This drives the edge of one plate beneath the other, so it slowly sinks into the hot mantle rock below the crust. Most of the places where this is happening are marked by deep trenches in the ocean floor, which trace the boundaries between the plates. These trenches can be more than twice the average depth of the oceans.

HOW TRENCHES ARE FORMED

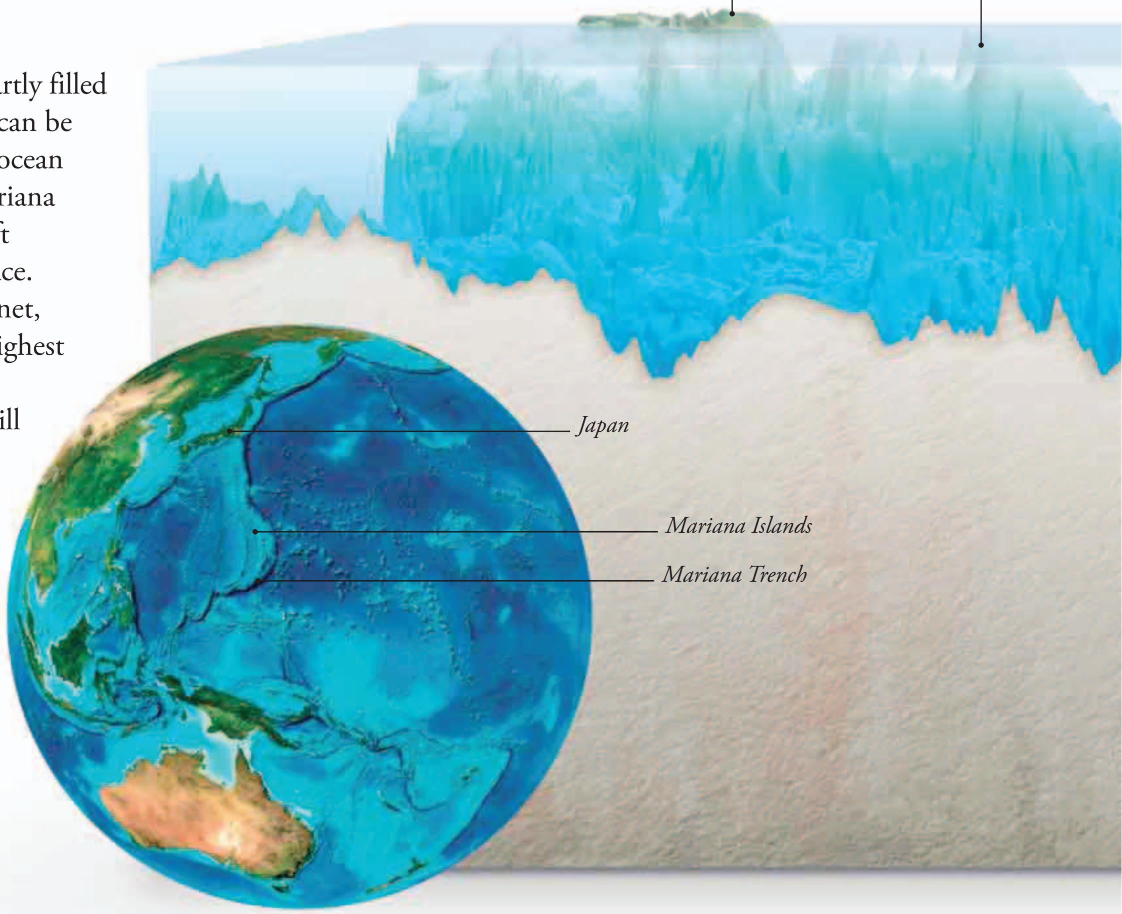
Deep ocean trenches lie above subduction zones, where part of the ocean floor is bending down to pass beneath another plate of Earth's crust. The steepest wall of the ocean trench is formed by the edge of the upper plate.



▲ DRAGGED UNDER
Heavy oceanic crust is always dragged beneath thicker but lighter continental crust, as shown here. Where both plates are oceanic, the oldest, heaviest plate slips beneath the younger one.

MARIANA TRENCH

Even though ocean trenches are partly filled with sand and muddy ooze, they can be three times as deep as the nearby ocean floor. The lowest point of the Mariana Trench in the Pacific lies 36,070 ft (10,994 m) below the ocean surface. It is the deepest chasm on the planet, deep enough to swallow Earth's highest mountain, Mount Everest in the Himalayas—and its peak would still be more than 6,560 ft (2,000 m) underwater. The subduction process not only formed this trench, it also created an arc of volcanic islands.



▸ CRESCENT-SHAPED
This image of the western Pacific shows the Mariana Trench as a dark line curving around the Mariana Islands. It is linked to other trenches extending north past Japan.